Fiscal Year:	FY 2007	**Task Last Updated:**	FY 03/26/2008
PI Name:	Ju, Yiguang Ph.D.		
Project Title:	STUDIES OF THERMAL RADIATION EFFECTS ON HIGH PRESSURE FLAME DYNAMICS		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	yju@princeton.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	609-258-5644
Organization Name:	Princeton University		
PI Address 1:	D329B, E-Quad, Olden Street		
PI Address 2:			
PI Web Page:			
City:	Princeton	**State:**	NJ
Zip Code:	08540-5263	**Congressional District:**	12
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	01/12/2004	**End Date:**	09/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Sacksteder, Kurt R	**Contact Phone:**	216-433-2857
Contact Email:	Kurt.R.Sacksteder@grc.nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 per PI (12/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC04GA59G		
Performance Goal No.:			
Performance Goal Text:			
	The effect of CO2 addition on flame speed and flammability limit of methane-air and methane-oxygen mixtures will be measured by a spherical propagating and counterflow flames. The spectral radiation will be detected and analyzed by using an infrared spectrometer and the InSb detector. An fitted statistical narrow band correlated-k (FSNB-CK)radiation model will be developed for spectral radiation prediction by comparing the predicted and measured radiation flux at given spectral wavelength. Task Objectives 1. Measure the effect of CO2 addition on flame speed and flammability limit of methane-air and methane-oxygen. 2. Develop a new flame initiation theory.		

Task Description:	3. Measure the the flame spectral radiation by using a spectrometer and InSb detector. 4. Develop an optimised radiation model 5. Design and build a drop module (rig). Task Significance: 1. Provide the first hand data of the effect of CO2 addition on fire control and burning limit. 2. Establish a new model for spectral radiation computation. 3. Provide flame radiation database for microgravity experiment 4. Established a new flame theory for fire suppression.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	1. Establish a simplified model for fire initation and extinguishment. 2. Enable modeling of fire spreading in microgravity and normal gravity enviroment. 3. Establish an accurate method for flame speed and flammability measurement. 4. Provide experimental data of CO2 radiation in flames and its impact on flammability limit. 5. Provide fire detection capability using a CO2 radiation sensor.
Task Progress:	1. The effects of spectral radiation absorption on flame speeds at normal and elevated pressures were experimentally and numerically investigated using the CO2 diluted outwardly propagating CH4-O2-He flames. Experimentally, the laminar burning velocities of CH4-O2-He-CO2 mixtures at both normal and elevated pressures (up to 5 atm) were measured by using a pressure-release type spherical bomb. The results showed that radiation absorption with CO2 addition increases the flame speed and extends the flammability limit. In addition, it was also shown that the increase of pressure augments the effect of radiation absorption. Computationally, a fitted statistical narrow-band correlated-k (FSNB-CK) model was developed and validated for accurate radiation prediction in spherical geometry. This new radiation scheme was integrated to the compressible flow solver developed to simulate outwardly propagating spherical flames (a software for simulations of unsteady reactive flow using adaptive grids (SURFA-1D)). The comparison between experiment and computation showed a very good agreement. The results showed that the flame geometry had a significant impact on radiation absorption and that the one-dimensional planar radiation model was not valid for the computation of the flame speed of a spherical flame. An effective Boltzmann number was extracted from numerical simulation. Furthermore, the FSNB-CK model was compared with the grey band SNB model. It was shown that the grey band SNB model over-predicts the radiation absorption. It is concluded that quantitative prediction of flame speed and flammability limit of CO2 diluted flame requires accurate spectral dependent radiation model. The results were reported on the 31st International Symposium on Combustion (2006). 2. Dynamics of flame kernel evolution with and without thermal radiation and external energy addition was investigated analytically and numerically. Emphasis was placed on the effects of radiation heat loss, ignition power, and Lewis number on the correlation and transition between the initial flame kernel, the self-extinguishing flame, the flame ball, the outwardly propagating spherical flame, and the propagating planar flame. The present study extended previous results by bridging the theories of the non-adiabatic stationary flame balls and traveling flames and allowing rigorous consideration of radiation heat losses. The results showed that the effects of radiation heat loss played an important role in flame regimes and flame transition and resulted in a new isolated self-extinguishing flame. Furthermore, it was found that radiation heat losses significantly increased the critical ignition radius and resulted in three different dependences of the minimum ignition power on the Lewis number. Comparisons between the transient numerical simulations and the quasi-steady state analytical results showed a good agreement. The results suggested that prediction of flame initiation without appropriate consideration of radiation is not acceptable. The results were published on Combustion Theory and Modeling (2006). 3. The combined effects of flame stretch, curvature, and radiation on the extinction and flammability limit of premixed tubular flames are investigated by using the large-activation-energy asymptotic method. An analytical expression for flame speed, flame temperature and extinction limits is obtained and used to study the combined effects on flame dynamics. The flame isola for near limit radiative tubular flames is predicted and compared with that of the counterflow planar flames. It is found that the flame curvature has great effects on both the stretch extinction limit and the radiation extinction limits. By varying the radiation intensity, the results showed that the combined effects of radiation and curvature lead to various flame bifurcations. Comparisons among tubular flames of different flame radii as well as counterflow flames, all at the same stretch rate and radiation intensity, indicate that the premixed flame can be strongly affected by flame curvature. Furthermore, the effects of Lewis number are examined. The results also demonstrate a significant influence of preferential diffusion on the extinction limits of tubular flames at different flame curvatures. These results are accepted for publication on the International Journal of Heat and Mass Transfer (2008). 4. The trajectories of outwardly propagating spherical flames initiated by an external energy deposition and radiation are studied theoretically, numerically, and experimentally by using hydrogen/air mixtures. Emphasis is placed on how to accurately determine the laminar flame speeds experimentally from the time history of flame fronts for mixtures with different Lewis numbers and ignition energies. The results show that there is a critical flame radius only above which are the linear and non-linear extrapolations for flame speeds valid. It is found that the critical radius depends strongly on the Lewis number. At large Lewis numbers, the critical radius is larger than the minimum flame radius used in the experimental measurements, leading to invalid flame speed extrapolation. The results also show that there is a maximum Karlovitz number beyond which propagating spherical flame does not exist. The maximum Karlovitz number decreases dramatically with the increase of Lewis number. Furthermore, the results show that the ignition energy has a significant impact on the flame trajectories. It is found that the unsteady flame transition causes a flame speed reverse phenomenon near the maximum Karlovitz number with different ignition energies. The occurrence of flame speed reverse greatly narrows the experimental data range for flame speed extrapolation. The strong dependence of flame trajectory on

ignition energy and the existence of flame speed reverse phenomenon are also confirmed by experimental results. The results will be reported at the 32nd International Symposium on Combustion (2008)

5. To accurately measure the flame speeds, the effects of flow compression and flame stretch on the accurate determination of laminar flame speeds at normal and elevated pressures using propagating spherical flames at constant pressure or constant volume are studied theoretically and numerically. The results show that both the compression-induced flow motion and flame stretch have significant impacts on the accuracy of flame speed determination. For the constant pressure method, a new method to obtain a Compression-Corrected Flame Speed (CCFS) for nearly constant pressure spherical bomb experiments is presented. Likewise, for the constant volume method, a technique to obtain a Stretch-Corrected Flame Speed (SCFS) at elevated pressures and temperatures is developed. The validity of theoretical results for both constant pressure and constant volume methods are demonstrated by numerical simulations using detailed chemistry for hydrogen/air, methane/air, and propane/air mixtures. It is shown that the present CCFS and SCFS methods not only improve the accuracy of the flame speed measurements significantly but also extend the parameter range of experimental conditions. The results can be used directly in experimental measurements of laminar flame speeds.

6. In order to experimentally validate the theory, the impact of thermal radiation and non-spherical flow geometry on the accuracy of flame speed measurement using a spherical combustion chamber is analyzed. The results showed that flame speed measurement based on the conventional assumptions without appropriate corrections of thermal radiation and volume confinement cause 10% or more errors. A new criterion for flame speed measurements using spherical bomb is presented. The results were presented at the 5th National Combustion Meeting (March, 2007) at UCSD and submitted for publication on Combustion and Flame (2008). Specifically, the effect of flow compression and stretch on flame speed measurements and the effect of non-spherical flow field on flame speed measurements were investigated experimentally. Effect of flow compression and stretch on flame speed measurements: Pressure waves produced in the experiments of flame speed measurement using a spherical bomb will induce an inward flow velocity in the burned gas region. This inward burned gas velocity will affect the measured flame speeds. Discussions of the compression effects on flame speed measurement were either missing or neglected in the previous studies. For the measured flame speed of a stoichiometric hydrogen-air mixture on the range of flame radii used in the experiments, the results show that due to the flow compression effect, the error of the measured flame speeds could be larger than 15% depending on range of flame radii chosen in the experiment. The correction of the flow compression effect by using an analytical expression significantly improved the accuracy of the flame speed measurements (less than 5%). To measure the flame speed at both high pressure and high temperature, a constant volume spherical flame bomb has been used. In this approach, the transient flame propagation speed was used directly as the flame speed of the mixture and the effect of flame stretch on the flame speed was neglected. The results also show that as the flame propagates outward, the flame speed obtained without flame stretch correction does not agree with the unstretched planar flame speed until the flame radius is larger than 60% of the chamber size. However, by considering the flame stretch effect, the agreement between the flame speeds computed from the planar flame and the spherical flame was improved significantly in a broader range of pressure and flame radius. Effect of non-spherical flow field on flame speed measurements: In addition to the compression and stretch effects, the non-spherical flow field in the cylindrical bomb also produced errors of flame speed measurement. An analytical relation to correct for the flow field effect in the cylindrical bomb was derived by using the potential flow analysis. The flame speeds of hydrogen/CO-air mixtures were measured at 1, 10, and 20 atm. The measured flame speeds were compared to the existing data and model predictions. For the comparison of the measured flame speeds with other experimental data and numerical predictions, the results show that the data reported in the literature are much lower than the present results. The comparison also shows that the present data agree well with the modeled results using the detailed hydrogen-CO mechanism. The results clearly demonstrate that the experimental accuracy of flame speed is a key issue for the validation of kinetic mechanism.

Bibliography Type:	Description: (Last Updated: 11/26/2025)
Articles in Peer-reviewed Journals	Chen Z, Ju Y. "Combined effects of curvature, radiation, and stretch on the extinction of premixed tubular flames." International Journal of Heat and Mass Transfer, in press, March 2008. Expected publication May 2008. , Mar-2008
Articles in Peer-reviewed Journals	Chen Z, Burke MP, Ju Y. "Effects of Lewis Number and Ignition Energy on the Determination of Laminar Flame Speed Using Propagating Spherical Flames." Proceedings of the Combustion Institute, Vol. 32, in press March 2008. Expected publication August 2008. , Mar-2008
Articles in Peer-reviewed Journals	Chen Z, Ju Y. "Theoretical analysis of the evolution from ignition kernel to flame ball and planar flame." Combustion Theory and Modeling, 2007 Jun;11(3):,427–53. http://dx.doi.org/10.1080/13647830600999850 , Jun-2007
Articles in Peer-reviewed Journals	Chen Z, Qin X, Xu B, Ju Y, Liu F. "Studies of radiation absorption on flame speed and flammability limit of CO2 diluted methane flames at elevated pressures." Proceedings of the Combustion Institute, 2007 Jan;31(2):2693-700. http://dx.doi.org/10.1016/j.proci.2006.07.202 , Jan-2007
Articles in Peer-reviewed Journals	Guo H, Smallwood GJ, Liu F, Ju Y, Gülder ÖL. "The effect of hydrogen addition on flammability limit and NOx emission in ultra-lean counterflow CH4/air premixed flames." Proceedings of the Combustion Institute 2005 Jan;30(1):303–11. http://dx.doi.org/10.1016/j.proci.2004.08.177 , Jan-2005
Articles in Peer-reviewed Journals	Ju Y, Xue Y. "Extinction and flame bifurcations of stretched dimethyl ether premixed flames." Proceedings of the Combustion Institute 2005 Jan;30(1):295–301. http://dx.doi.org/10.1016/j.proci.2004.08.258 , Jan-2005
Articles in Peer-reviewed Journals	Burke MP, Chen Z, Ju Y, Dryer FL. "On the Determination of Laminar Flame Speed by Using Outwardly Propagating Flames: Effect of Cylindrical Confinement." Combustion and Flame, submitted, 2008. , Jan-2008
Papers from Meeting Proceedings	Burke MP, Chen Z, Ju Y, Dryer FL. "On the Determination of Laminar Flame Speed by Using Outwardly Propagating Flames: Effect of Cylindrical Confinement." 5th National Combustion Meeting, University of California San Diego, March 2007. 5th National Combustion Meeting, University of California San Diego, March 2007. , Mar-2007